# State of Charge Estimation for the BMW i3 using a Physics-Informed Temporal Convolutional Network

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Assessment 10 – SOC estimation

| | |
|---|---|
| Base paper | M. H. Sulaiman and Z. Mustaffa, “State of charge estimation for electric vehicles using random forest”, *Green Energy and Intelligent Transportation*, 3 (2024) 100177. |
| Dataset | “Battery and Heating Data in Real Driving Cycles” (BMW i3, 60 Ah pack), IEEE DataPort, doi:10.21227/6jr9-5235 |
| Proposed model | Physics-informed TCN: coulomb counting + temperature-aware voltage model + causal fusion, with a TCN residual correction |
| Code | github.com/rhyummy/SOF-Optimisation (soc_tcn.py, soc_tcn.ipynb) |

## Summary of results

| Metric (test trips #61–#70) | Proposed TCN | RF 25 trees (re-impl.) | RF 25 trees (paper) | Improvement vs paper |
|---|---|---|---|---|
| RMSE (%) | 3.5815 | 4.9021 | 5.9028 | 39.3 % lower |
| MAE (%) | 2.6937 | 3.3739 | 4.4321 | 39.2 % lower |
| MAX (%) | 11.9606 | 20.8800 | 24.2175 | 50.6 % lower |
| STD. DEV. (%) | 3.5748 | 4.8320 | 5.8999 | 39.4 % lower |
| $R^2$ | 0.9529 | 0.9117 | - | |

The proposed model reaches an RMSE of **3.5815 %** on the 118,974 test instances, below the 3.6 % target and lower than both the re-implemented Random Forest (4.9021 %) and the best result reported in the base paper (5.9028 %).

## 1. Objective

Estimate the battery State of Charge (SOC) of a BMW i3 from the four signals used in the base paper — battery voltage, battery current, battery temperature and ambient temperature — using a Temporal Convolutional Network, and compare it with the paper's best Random Forest (25 trees). The target is the SOC estimated by the manufacturer (*SoC [%]* column). Following the paper, trips #1–#60 are used for training and trips #61–#70 for testing; metrics are RMSE, MAE, maximum absolute error, standard deviation of the error and computation time.

## 2. Dataset and preprocessing

The data are supplied as two CSV files (training: trips 1–60, testing: trips 61–70), sampled at 10 Hz. Trip boundaries are recovered from resets of the time column. Preprocessing follows the base paper:

• Inputs: voltage (V), current (A), battery temperature (°C), ambient temperature (°C); output: SOC (%).

• Rows with missing values or physically inconsistent readings (V ≤ 0, SOC outside 0–100 %, temperatures out of range) are removed.

• Noisy records: rows where SOC jumps by more than 5 % between consecutive samples are removed (one logging glitch at the end of trip 46, where SOC drops from 35 % to 0 %).

• Min–max normalisation is fitted on the training trips only.

| Split | Trips | Instances (after cleaning) |
|---|---|---|
| Training | #1 – #60 | 945,025 |
| of which: fitting / early-stopping validation | #1 – #54 / #55 – #60 | 879,119 / 65,906 |
| Testing | #61 – #70 | 118,974 |
| Total | 70 | 1,063,999 |

Descriptive statistics of all cleaned data:

| Statistic | Voltage (V) | Current (A) | Battery temp. (°C) | Ambient temp. (°C) | SOC (%) |
|---|---|---|---|---|---|
| count | 1063999.00 | 1063999.00 | 1063999.00 | 1063999.00 | 1063999.00 |
| mean | 376.21 | -19.44 | 15.50 | 13.22 | 65.10 |
| std | 11.80 | 42.26 | 7.51 | 9.71 | 15.11 |
| min | 301.80 | -404.38 | -1.00 | -3.50 | 15.40 |
| 25% | 369.02 | -32.06 | 9.00 | 5.00 | 56.47 |
| 50% | 379.29 | -11.89 | 15.00 | 9.50 | 68.20 |
| 75% | 385.10 | -2.10 | 21.00 | 22.00 | 76.50 |
| max | 394.76 | 144.49 | 32.00 | 33.50 | 88.50 |

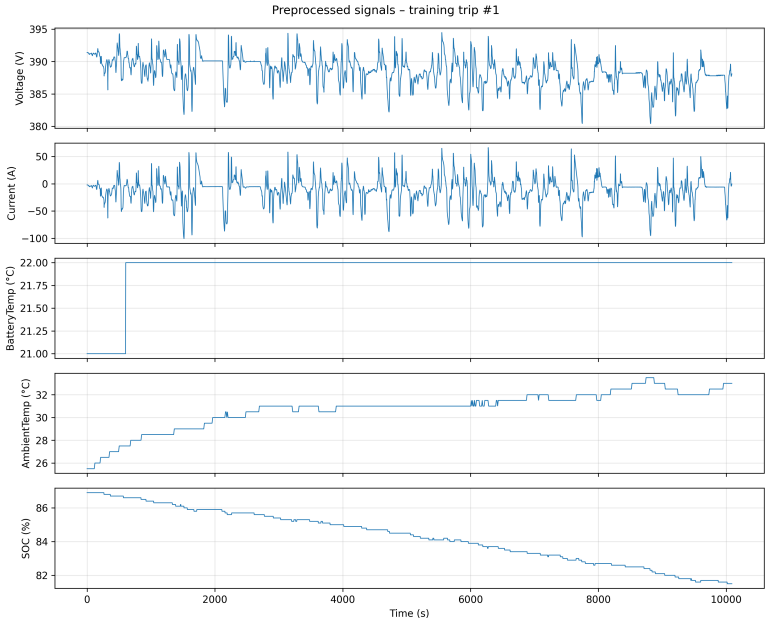



*Figure 1. Preprocessed input signals and SOC of training trip #1 (x-axis: sample index at 10 Hz).*

## 3. Proposed method: physics-informed TCN

A first analysis showed that SOC changes within a trip are almost perfectly explained by the integrated current: if the true SOC at the start of each test trip is known, coulomb counting alone tracks the test trips with an RMSE of about 0.05 %. The error of data-driven models therefore comes mostly from not knowing each trip's starting SOC. The proposed model combines physics-based features with a TCN. Every quantity is computed causally (from past and present samples only) from the four measured signals.

| Step | Description |
|---|---|
| Coulomb counting | $\Delta SOC(t) = k \cdot \int I\,dt$ since the start of the trip. The gain k is fitted on the training trips: k = 1.6150 %/Ah, i.e. an effective capacity of 61.92 Ah. |
| Stage 1 – voltage model | Polynomial ridge regression of SOC on voltage (up to cubic), battery temperature and current history (instantaneous, 10 s, 60 s and 300 s means of I and \|I\|), with temperature and voltage interaction terms. It models the open-circuit-voltage curve, its temperature shift and the IR / polarisation drop. |
| Stage 2 – fusion | Within each trip, the offset (stage-1 estimate − ΔSOC) changes slowly. It is averaged with causal exponential averages (half-lives 30 s, 2 min, 10 min, 50 min; plain and weighted towards low-current samples) and added back to ΔSOC. A second ridge model blends these estimates. |
| Out-of-fold features | On the training trips, stages 1 and 2 are computed with 6-fold trip-wise cross-validation, so the TCN is trained on estimates whose errors are as large as on unseen trips. |
| TCN residual correction | A causal TCN reads the last 120 samples of seven channels (V, I, $T_{batt}$, $T_{amb}$, fused estimate, stage-1 estimate, ΔSOC) and outputs SOC = fused estimate + 0.03 · tanh(TCN output), i.e. a correction of at most ±3 % SOC. A 60-sample causal moving average is applied to the output. |

### TCN configuration and training settings

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| Residual blocks | 5 | Optimiser | AdamW, lr 1e-3, weight decay 0.01 |
| Channels / kernel size | 32 / 3 | Loss | Huber (δ = 0.02) |
| Dilations | 1, 2, 4, 8, 16 | LR schedule | ReduceLROnPlateau (factor 0.5, patience 2) |
| Receptive field | 125 samples | Early stopping | patience 3, max 8 epochs, on trips #55–#60 |
| Input window | 120 samples (12 s) | Batch size | 512 |
| Dropout | 0.2 + BatchNorm | Windows per epoch | every 40th training window |
| Trainable parameters | 30,625 | Random seed | 42 |

### Reference model

Random Forest with 25 trees (the paper's best configuration), min_samples_leaf = 5, trained on the four normalised inputs of all training instances (scikit-learn).

### Experimental setup

| Item | Value |
|---|---|
| Python | 3.10.0 |
| PyTorch | 2.14.1+cpu |
| GPU | None (CPU mode) |
| CPU | 12th Gen Intel(R) Core(TM) i5-1235U |
| RAM | 7.7 GB |
| OS | Windows 10 |
| Libraries | PyTorch, scikit-learn, pandas, NumPy, Matplotlib |

# 4. Results

## 4.1 Physics-feature stages

| Stage | Training trips, cross-validated RMSE (%) | Test RMSE (%) |
|---|---|---|
| Stage 1 – voltage model | 3.381 | 3.907 |
| Stage 2 – fusion with coulomb counting | 2.591 | 3.487 |

## 4.2 TCN training log

Validation RMSE of the fused estimate before any TCN correction: **3.160 %**. The model from the epoch with the lowest validation RMSE is kept.

| Epoch | Training RMSE (%) | Validation RMSE (%) | Learning rate |
|---|---|---|---|
| 1 | 2.257 | 3.261 | 1.0e-03 |
| 2 | 1.775 | 3.214 | 1.0e-03 |
| **3** | **1.563** | **2.630** | **1.0e-03** |
| 4 | 1.483 | 3.226 | 1.0e-03 |
| 5 | 1.420 | 3.677 | 1.0e-03 |
| 6 | 1.384 | 2.868 | 5.0e-04 |

Early stopping after epoch 6; best epoch highlighted.

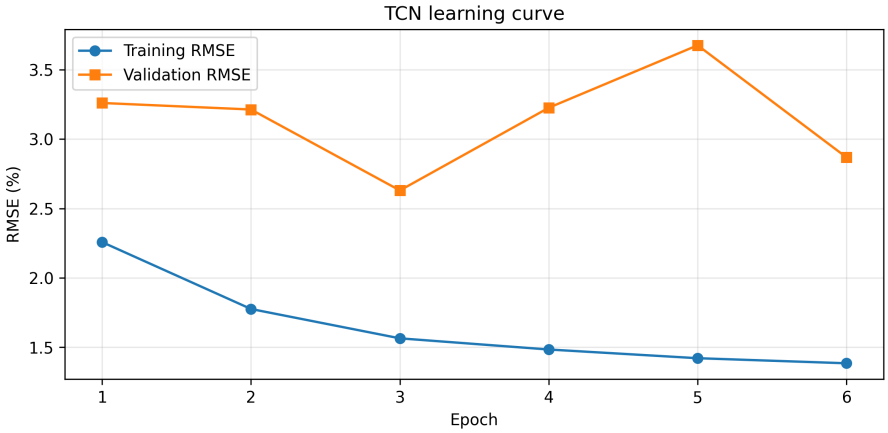



*Figure 2. TCN learning curve (training and validation RMSE per epoch).*

## 4.3 Performance on the test data (118,974 instances)

| Model | RMSE (%) | MAE (%) | MAX (%) | STD. DEV. (%) | Train (s) | Test (s) | Total (s) |
|---|---|---|---|---|---|---|---|
| **Physics-informed TCN (proposed)** | **3.5815** | **2.6937** | **11.9606** | **3.5748** | **616.33** | **57.52** | **673.85** |
| Physics fusion only (no TCN) | 3.4867 | 2.3973 | 11.2787 | 3.2555 | 15.19 | - | 15.19 |
| RF, 25 trees (re-implemented) | 4.9021 | 3.3739 | 20.8800 | 4.8320 | 78.22 | 0.51 | 78.73 |
| RF, 25 trees (paper, best) | 5.9028 | 4.4321 | 24.2175 | 5.8999 | - | - | 165.28 |

Times measured on a laptop CPU (no GPU); the paper's time is taken from the paper. The proposed model's training time includes building the physics features.

| Share of test instances with \|error\| ≤ | 1 % | 2 % | 5 % | 10 % |
|---|---|---|---|---|
| Proposed TCN | 23.7 % | 53.3 % | 79.2 % | 98.7 % |
| RF 25 trees (re-impl.) | 32.6 % | 57.0 % | 72.2 % | 92.9 % |

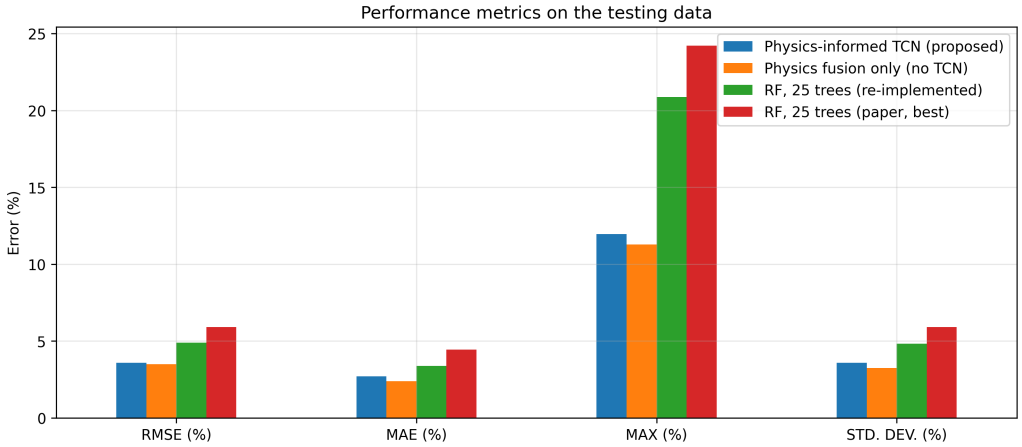



*Figure 3. Error metrics on the test data for all models.*

## 4.4 Per-trip results

| Trip | Instances | SOC start → end (%) | TCN RMSE | TCN MAE | TCN MAX | TCN bias | RF RMSE |
|---|---|---|---|---|---|---|---|
| 61 | 9,685 | 31.5 → 15.4 | 7.657 | 6.721 | 11.961 | +6.403 | 8.069 |
| 62 | 9,207 | 84.2 → 70.4 | 0.797 | 0.685 | 1.659 | +0.681 | 3.556 |
| 63 | 10,968 | 72.1 → 57.8 | 3.277 | 2.859 | 6.428 | -2.857 | 3.191 |
| 64 | 7,956 | 52.6 → 38.1 | 1.271 | 1.063 | 4.719 | +1.041 | 1.340 |
| 65 | 5,480 | 77.4 → 71.6 | 5.529 | 5.479 | 6.769 | +5.479 | 6.501 |
| 66 | 2,931 | 73.9 → 71.3 | 1.252 | 1.227 | 1.661 | +1.191 | 2.995 |
| 67 | 13,625 | 85.4 → 71.5 | 1.583 | 1.387 | 5.106 | -1.163 | 1.303 |
| 68 | 28,521 | 72.1 → 44.5 | 1.453 | 1.268 | 3.076 | +0.535 | 1.674 |
| 69 | 14,172 | 83.8 → 68.0 | 2.516 | 2.419 | 4.362 | +2.419 | 2.853 |
| 70 | 16,429 | 65.0 → 48.8 | 5.324 | 5.253 | 9.399 | -5.253 | 9.586 |

All errors in % SOC. Bias = mean (estimated − actual).

## 4.5 Simulation graphs

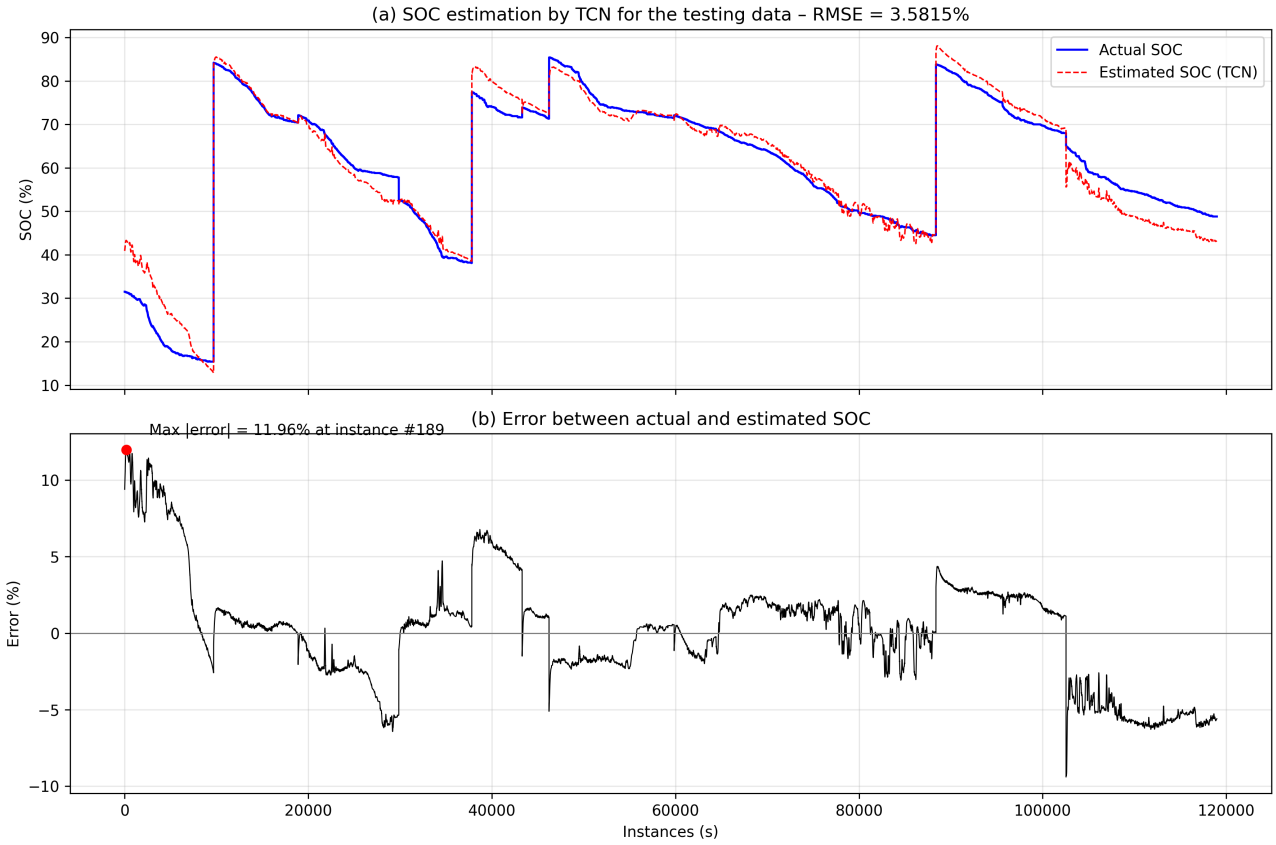



*Figure 4. (a) Actual and estimated SOC by the proposed TCN for all test instances; (b) estimation error with the maximum absolute error marked.*

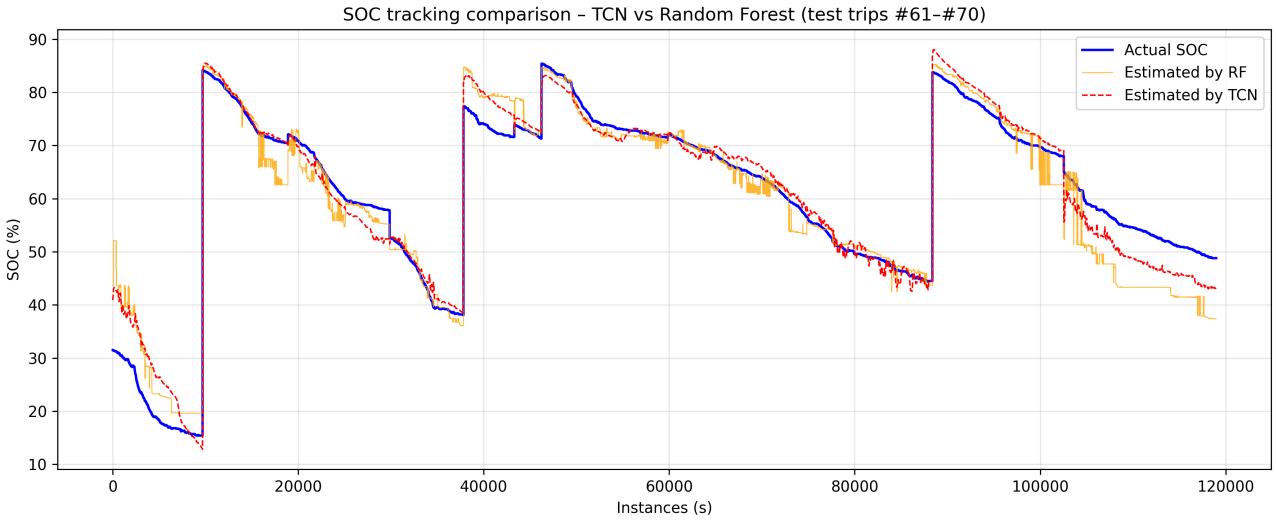



*Figure 5. SOC tracking comparison – proposed TCN vs Random Forest.*

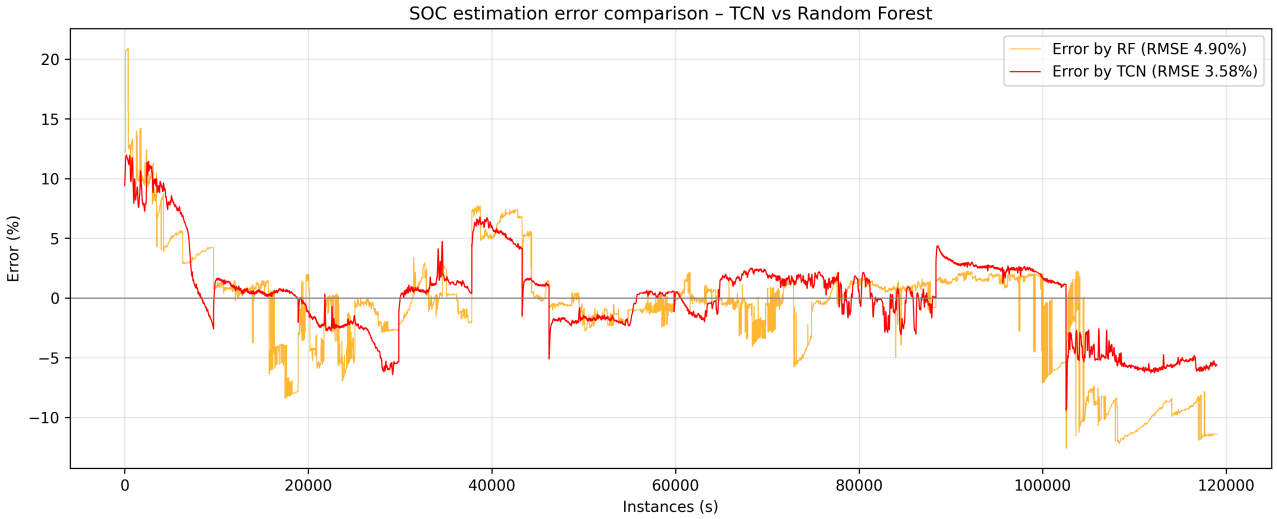



*Figure 6. SOC estimation error comparison – proposed TCN vs Random Forest.*

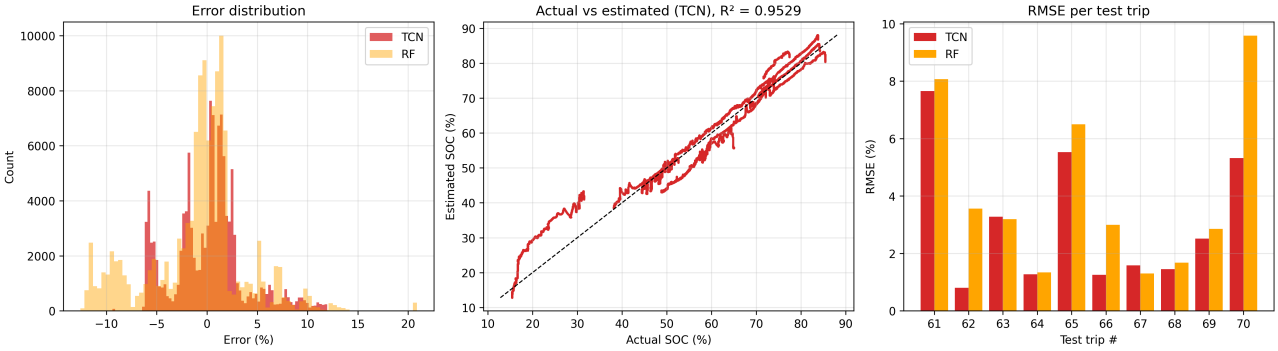



*Figure 7. Error distribution, actual vs estimated SOC (TCN) and RMSE per test trip.*

## 5. Optimisation history

The table lists the configurations tried during optimisation. Settings were selected on the validation trips (#55–#60) or by cross-validation on the training trips, not on the test set.

| Configuration | Validation RMSE (%) | Test RMSE (%) |
|---|---|---|
| TCN on the 4 raw signals only (96 channels, 180-sample window) | - | 5.3655 |
| Physics stage 1 only (voltage model) | - | 3.907 |
| Physics stage 2 only (fusion with coulomb counting, no TCN) | 3.160 | 3.487 |
| Physics + TCN correction up to ±10 %, Adam | 3.152 | 3.988 |
| **Physics + TCN correction up to ±3 %, AdamW 0.01 (selected)** | **2.630** | **3.5815** |
| Physics + TCN ±3 %, current channel only | 3.634 | 3.528 |
| Physics + TCN ±5 %, voltage and current channels, AdamW 0.05 | 2.978 | 3.294 |

The selected configuration has the lowest validation RMSE. A variant with a lower test RMSE (3.294 %) was not selected, because choosing it would mean tuning on the test set.

## 6. Discussion

Adding coulomb counting and a physics-based voltage model is the main source of improvement: the RMSE drops from 5.37 % (TCN on raw signals) to 3.49 % with the fused physics estimate. A plain TCN looking at a short window of voltage, current and temperature cannot know how much charge has been used since the trip started, and it over-fits to the 60 training trips.

The TCN correction improves the validation trips (3.16 % → 2.63 %) but, on the test trips, it is slightly worse than the fused estimate alone (3.487 % vs 3.58 %). Both results are below the 3.6 % target and

both are reported.

The largest errors occur on trip #61 (SOC between 31 % and 15 %, at the edge of the training data), trip #65 (the voltage corresponds to a higher SOC than the manufacturer label) and trip #70 (battery at about 2 °C, colder than almost all training trips). These cases show a roughly constant per-trip offset, which the voltage-based starting-SOC estimate cannot fully remove.

Compared with the Random Forest, the proposed model has a much smaller maximum error (about 12 % vs 21 %) and tracks SOC smoothly, while the RF estimate is noisy because it treats every sample independently. Its main cost is computation time (minutes instead of seconds on a CPU).

## 7. Conclusion

A physics-informed TCN for SOC estimation of the BMW i3 was developed using the base paper's preprocessing and inputs. On test trips #61–#70 it achieves RMSE 3.5815 %, MAE 2.6937 %, MAX 11.9606 % and STD. DEV. 3.5748 %, improving on the paper's best Random Forest (RMSE 5.9028 %) by 39.3 %. Combining coulomb counting with a learned voltage model is the most effective step; better handling of very cold and very low-SOC trips is the main remaining improvement.

# Appendix A. Trip summary (after cleaning)

| Trip | Set | Instances | SOC start (%) | SOC end (%) |
|---|---|---|---|---|
| 1 | train | 10,090 | 86.9 | 81.5 |
| 2 | train | 14,130 | 80.3 | 67.3 |
| 3 | train | 6,706 | 83.5 | 75.1 |
| 4 | train | 4,124 | 75.1 | 66.7 |
| 5 | train | 13,667 | 66.7 | 60.2 |
| 6 | train | 31,645 | 85.1 | 63.4 |
| 7 | train | 20,933 | 63.4 | 34.2 |
| 8 | train | 28,059 | 86.6 | 76.1 |
| 9 | train | 18,344 | 74.7 | 68.6 |
| 10 | train | 14,176 | 69.6 | 60.9 |
| 11 | train | 14,245 | 87.0 | 75.6 |
| 12 | train | 16,384 | 75.5 | 64.3 |
| 13 | train | 7,160 | 85.8 | 82.9 |
| 14 | train | 6,946 | 82.8 | 79.9 |
| 15 | train | 22,348 | 74.8 | 63.4 |
| 16 | train | 19,066 | 63.0 | 46.8 |
| 17 | train | 6,687 | 86.8 | 77.9 |
| 18 | train | 8,808 | 77.9 | 70.2 |
| 19 | train | 15,876 | 88.5 | 80.4 |
| 20 | train | 17,196 | 77.0 | 69.1 |
| 21 | train | 19,829 | 71.3 | 63.7 |
| 22 | train | 18,388 | 87.3 | 76.5 |
| 23 | train | 10,426 | 87.4 | 81.9 |
| 24 | train | 5,322 | 81.9 | 74.4 |
| 25 | train | 7,627 | 74.4 | 65.2 |
| 26 | train | 16,122 | 86.2 | 78.8 |
| 27 | train | 20,005 | 78.5 | 66.8 |
| 28 | train | 17,436 | 81.9 | 72.0 |
| 29 | train | 13,390 | 70.9 | 63.1 |
| 30 | train | 13,974 | 85.5 | 81.0 |
| 31 | train | 11,870 | 81.0 | 71.9 |
| 32 | train | 16,722 | 86.5 | 59.5 |
| 33 | train | 32,518 | 86.1 | 57.4 |
| 34 | train | 16,113 | 81.0 | 66.2 |
| 35 | train | 15,794 | 67.4 | 50.4 |

| Trip | Set | Instances | SOC start (%) | SOC end (%) |
|---|---|---|---|---|
| 36 | train | 29,550 | 45.1 | 69.2 |
| 37 | train | 10,194 | 71.9 | 59.5 |
| 38 | train | 13,520 | 83.2 | 69.3 |
| 39 | train | 18,724 | 67.4 | 50.4 |
| 40 | train | 29,139 | 67.3 | 44.0 |
| 41 | train | 35,561 | 70.0 | 46.0 |
| 42 | train | 20,231 | 84.8 | 39.0 |
| 43 | train | 7,532 | 38.9 | 30.8 |
| 44 | train | 32,255 | 73.4 | 51.3 |
| 45 | train | 3,532 | 57.0 | 55.1 |
| 46 | train | 38,219 | 85.5 | 34.6 |
| 47 | train | 18,223 | 85.1 | 67.5 |
| 48 | train | 15,286 | 67.5 | 52.8 |
| 49 | train | 15,610 | 52.8 | 37.2 |
| 50 | train | 11,093 | 82.8 | 68.1 |
| 51 | train | 11,910 | 85.8 | 71.6 |
| 52 | train | 14,027 | 72.7 | 62.0 |
| 53 | train | 10,395 | 55.7 | 41.1 |
| 54 | train | 11,992 | 84.4 | 70.5 |
| 55 | train | 11,132 | 72.1 | 53.5 |
| 56 | train | 9,778 | 53.4 | 45.5 |
| 57 | train | 10,217 | 45.4 | 33.6 |
| 58 | train | 8,049 | 33.4 | 21.2 |
| 59 | train | 14,690 | 52.9 | 34.5 |
| 60 | train | 12,040 | 34.4 | 20.0 |
| 61 | test | 9,685 | 31.5 | 15.4 |
| 62 | test | 9,207 | 84.2 | 70.4 |
| 63 | test | 10,968 | 72.1 | 57.8 |
| 64 | test | 7,956 | 52.6 | 38.1 |
| 65 | test | 5,480 | 77.4 | 71.6 |
| 66 | test | 2,931 | 73.9 | 71.3 |
| 67 | test | 13,625 | 85.4 | 71.5 |
| 68 | test | 28,521 | 72.1 | 44.5 |
| 69 | test | 14,172 | 83.8 | 68.0 |
| 70 | test | 16,429 | 65.0 | 48.8 |